Block Endocrine 23 July 2025 – Year 2 Med Student

Obesity

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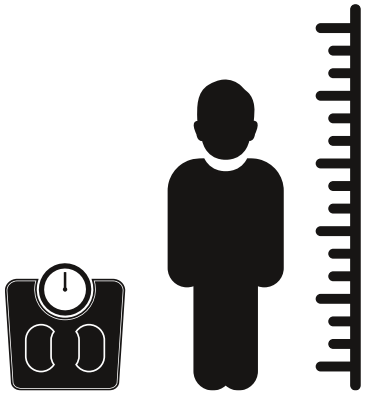
คุณจะรู้ได้ยังงัยว่าคุณ “อ้วน” ?





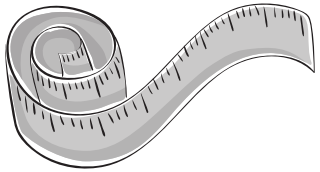
BMI

Asia Pacific population



<18.5	underweight
18.5-22.9	normal
23-24.9	overweight
>/=25	obesity

WC



Male >90 cm

Female >80 cm

Definition and classification of obesity

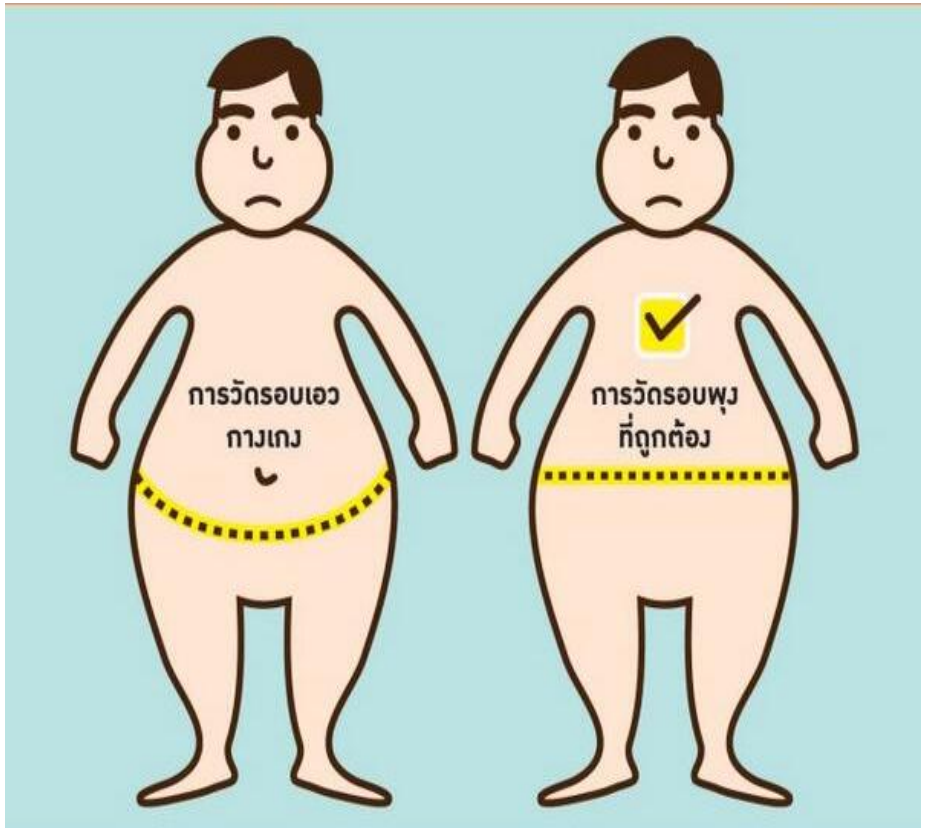
- Obesity is defined as abnormal or excessive fat accumulation that may impair health
- Body mass index (BMI) provides the most convenient population-level measure of overweight and obesity currently available

$$BMI = \frac{weight\ (kg)}{height^2\ (m^2)}$$

Classification	BMI (kg/m²)		
	International classification[1]	Asian population[2]	Japanese guidelines[3]
Underweight	<18.5		<18.5
Normal range	≥18.5 and <25	≥18 and <23	≥18.5 - <25
Pre-obesity*	≥25 and <30	≥23 and <25	
Obesity	≥30	>25	
Obesity class I	≥30 and <35		≥25 and <30
Obesity class II	≥35 and <40		≥30 and <35
Obesity class III	≥40		≥35 and <40
Obesity class IV			≥40

*Previously described as overweight according to WHO nomenclature.
BMI, body mass index; JASSO, Japan Society for the Study of Obesity; WHO, World Health Organization.
1. WHO. Obesity: preventing and managing the global epidemic. 2000. Available from https://www.who.int/nutrition/publications/obesity/WHO_TRS_894/en/. Accessed May 2020;
2. Misra A et al. J Assoc Physicians India. 2009;57:163–70; 3. Guidelines for the management of obesity disease 2016 (Japan), issued by JASSO.

Waist to height ratio > 0.5





Asia Pacific Journal of Public HealthVolume 34, Issue 1, January 2022, Pages 36-43, Article Reuse Guidelines https://doi.org/10.1177/10105395211046474

BMI (Body Mass Index) = Body weight (kg) / Height (m^2)



Same Height

Same Weight

Same BMI

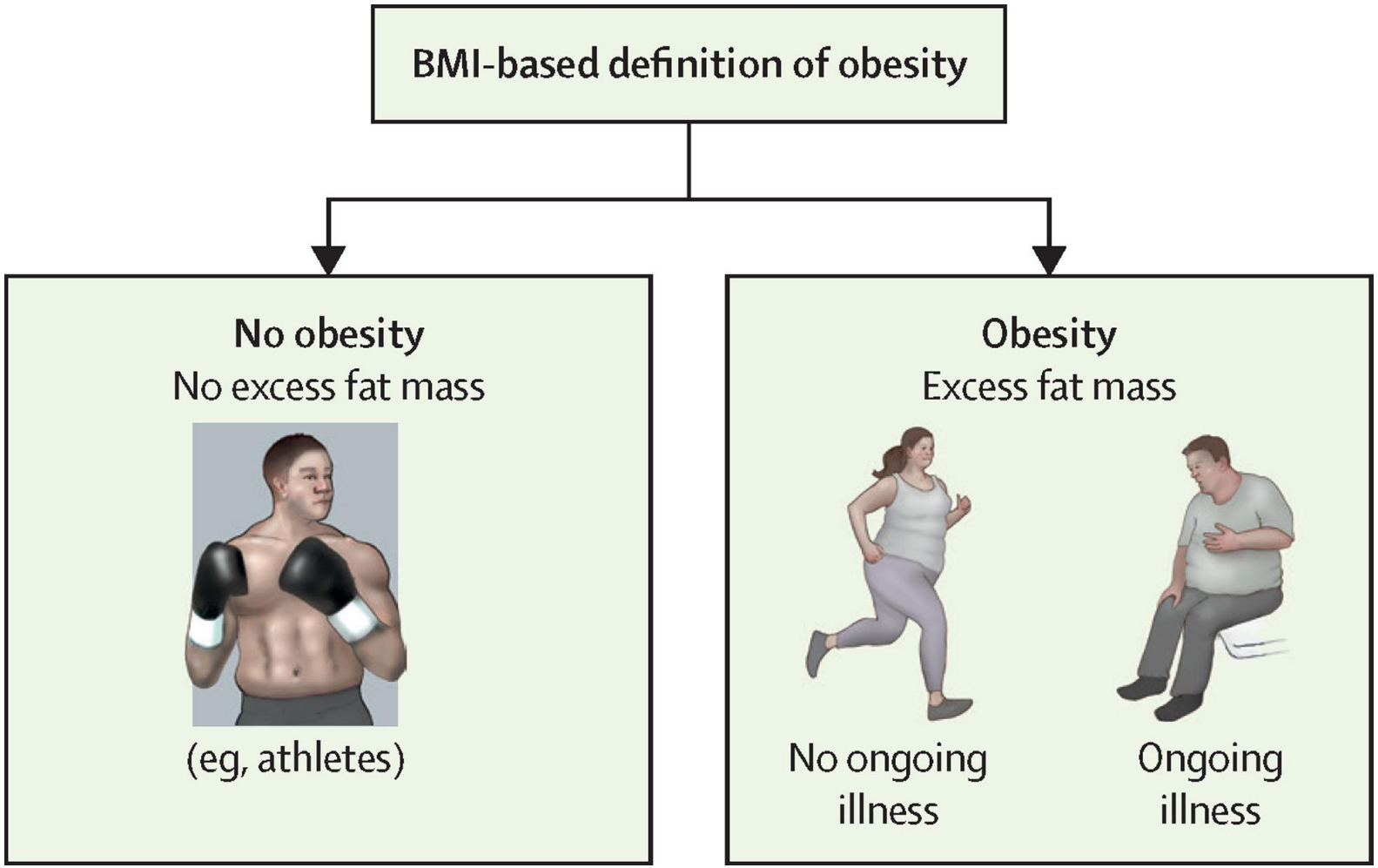


70 kg

70 kg

Different Body Fat & Muscle Proportion

Limitations of the current framing of obesity



Rubino, Francesco et al. The Lancet Diabetes & Endocrinology, Volume 13, Issue 3, 221 - 262

Diagnosing clinical obesity

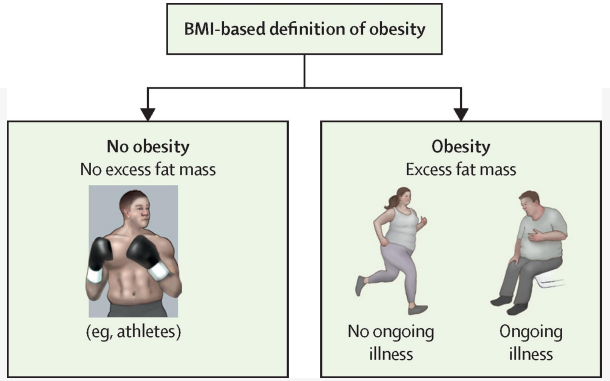
Limitations of the current definition of obesity

Obesity is currently defined solely by an individual's body mass index (BMI)

The criteria for populations of European descent* are:

Diagnosis	Underweight	Normal	Overweight	Obesity
BMI (kg/m²)	Under 18.5	18.5 to 24.9	25 to 29.9	30 and over

Criteria for other ethnic groups are different



- ✓ Although BMI is **useful** for identifying individuals at increased risk of health consequences...
- ✗ It **is not** a direct measure of fat
- ✗ It **does not** establish the distribution of fat around the body
- ✗ It **cannot** determine when excess body fat is a health problem

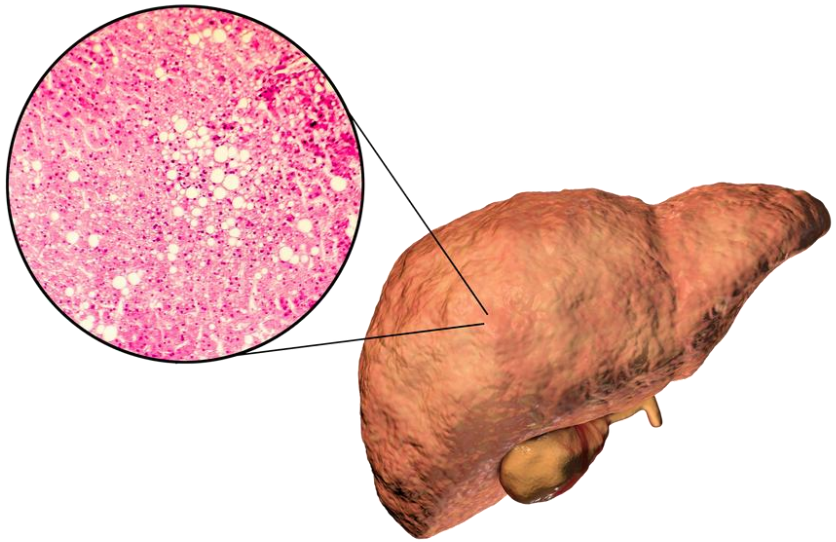
Rubino, Francesco et al. The Lancet Diabetes & Endocrinology, Volume 13, Issue 3, 221 - 262

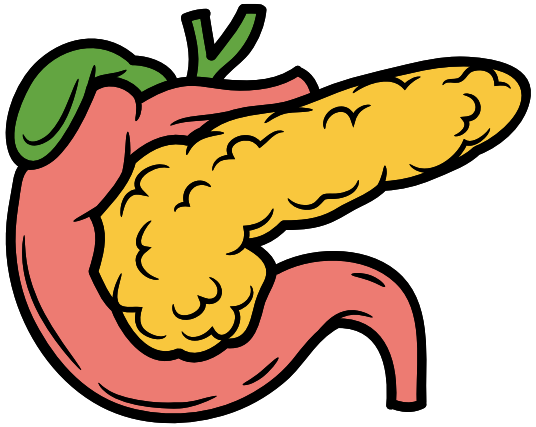
Obesity

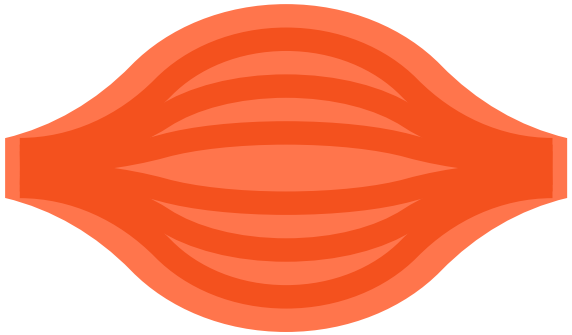
a chronic disease

abnormal or excessive fat accumulation
that presents a risk to health









Elements of disease

Aetiology → Pathophysiology →  → Illness

Cause of disease

Specific disease mechanisms

Compromised organ function

Clinical manifestations

Traditional framing of obesity

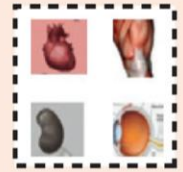
Aetiology → Pathophysiology → Obesity-related diseases (eg, diabetes and cancer)

Additional aetiologies or pathophysiology → Obesity-related diseases (eg, diabetes and cancer)

Pathophysiology ⇢ Direct effects of excess adiposity

(eg, genetic, environmental, or unknown)

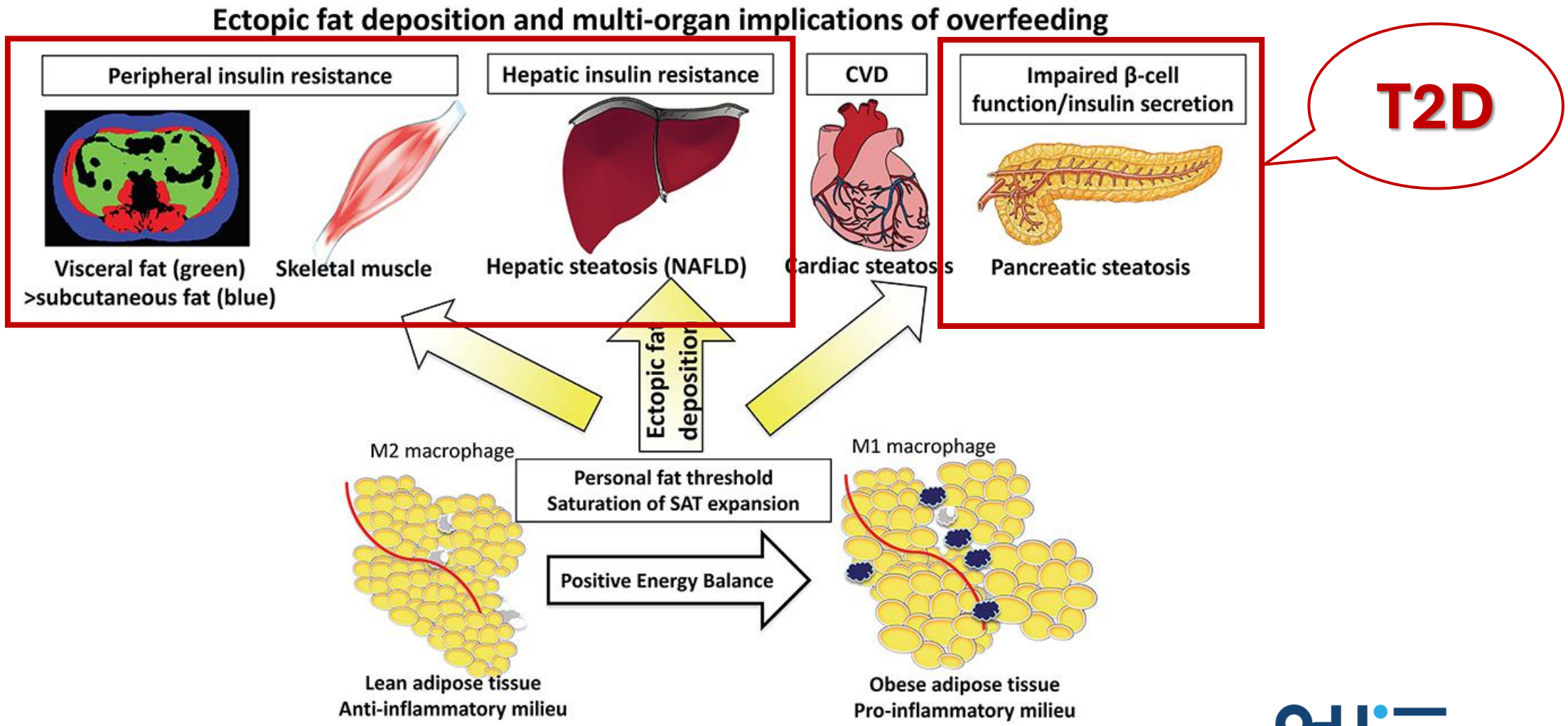
(eg, mechanical, endocrine, metabolic, or inflammatory)

Direct effects of excess adiposity:  ⇢ Obesity as an illness?

Compromised organ function(?)

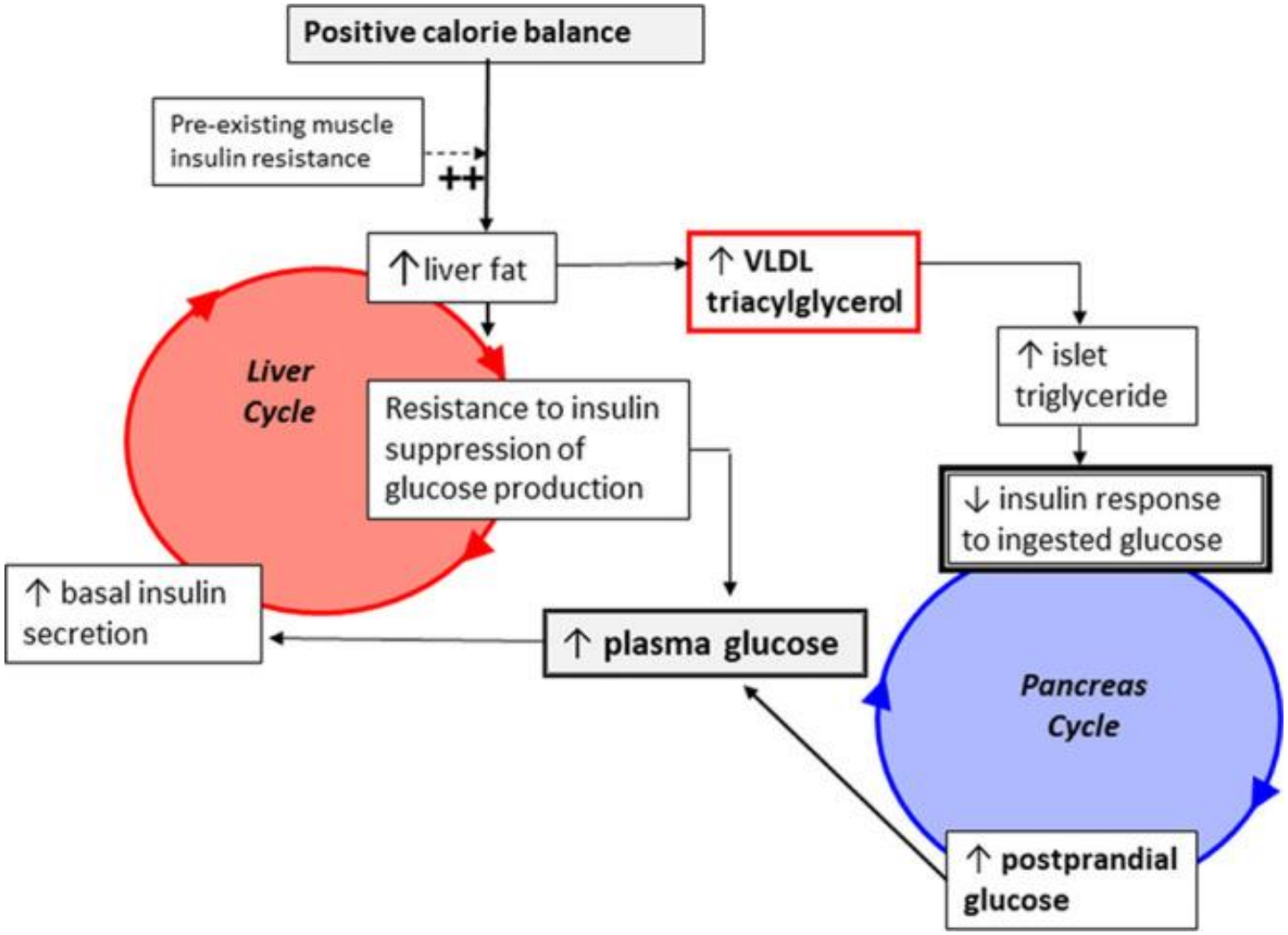
Clinical manifestations

Excessive energy intake is a key driver of metabolic diseases



The Impact of Macronutrient Intake on Non-alcoholic Fatty Liver Disease (NAFLD): Too Much Fat, Too Much Carbohydrate, or Just Too Many Calories? Front. Nutr.2021. 8:640557.

The twin cycle hypothesis of the aetiology of T2D



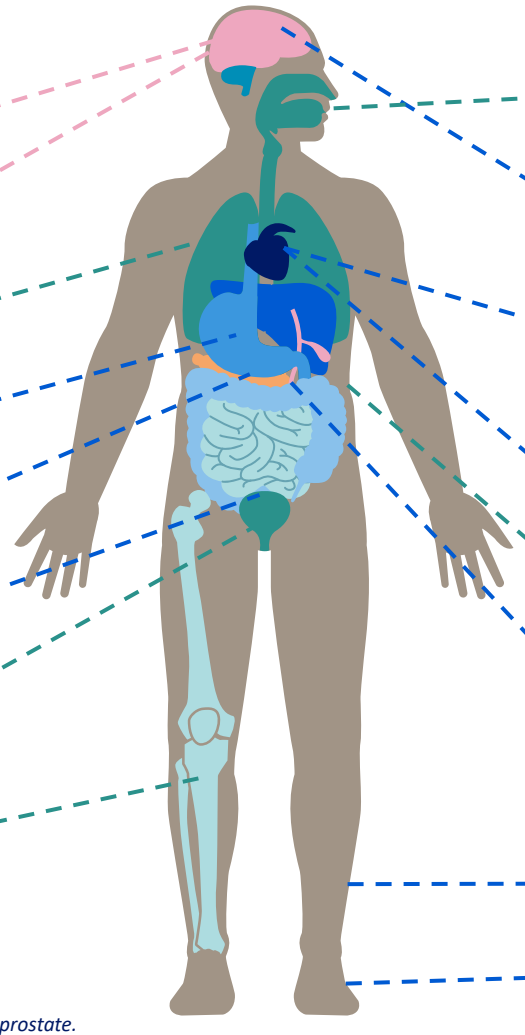
Obesity is associated with multiple complications including CVD

Metabolic

Mechanical

Mental

- GERD
- Cancers*
- Physical functioning
- Depression
- Anxiety
- Asthma
- MAFLD
- Gallstones
- Infertility
- Incontinence
- Knee osteoarthritis



- Sleep apnoea
- **Cardiovascular diseases**
  - Stroke
  - Dyslipidaemia
  - Hypertension
  - Coronary artery disease
  - Pulmonary embolism
  - HFpEF
- Coronary heart disease
- Chronic back pain
- T2D
  Prediabetes
- Thrombosis
- Gout

**Including postmenopausal breast, colorectal, endometrial, oesophageal, kidney, ovarian, pancreatic and prostate.*

CVD, cardiovascular disease; GERD, gastro-oesophageal reflux disease; HFpEF, heart failure with preserved ejection fraction; NAFLD, non-alcoholic fatty liver disease; T2D, type 2 diabetes.

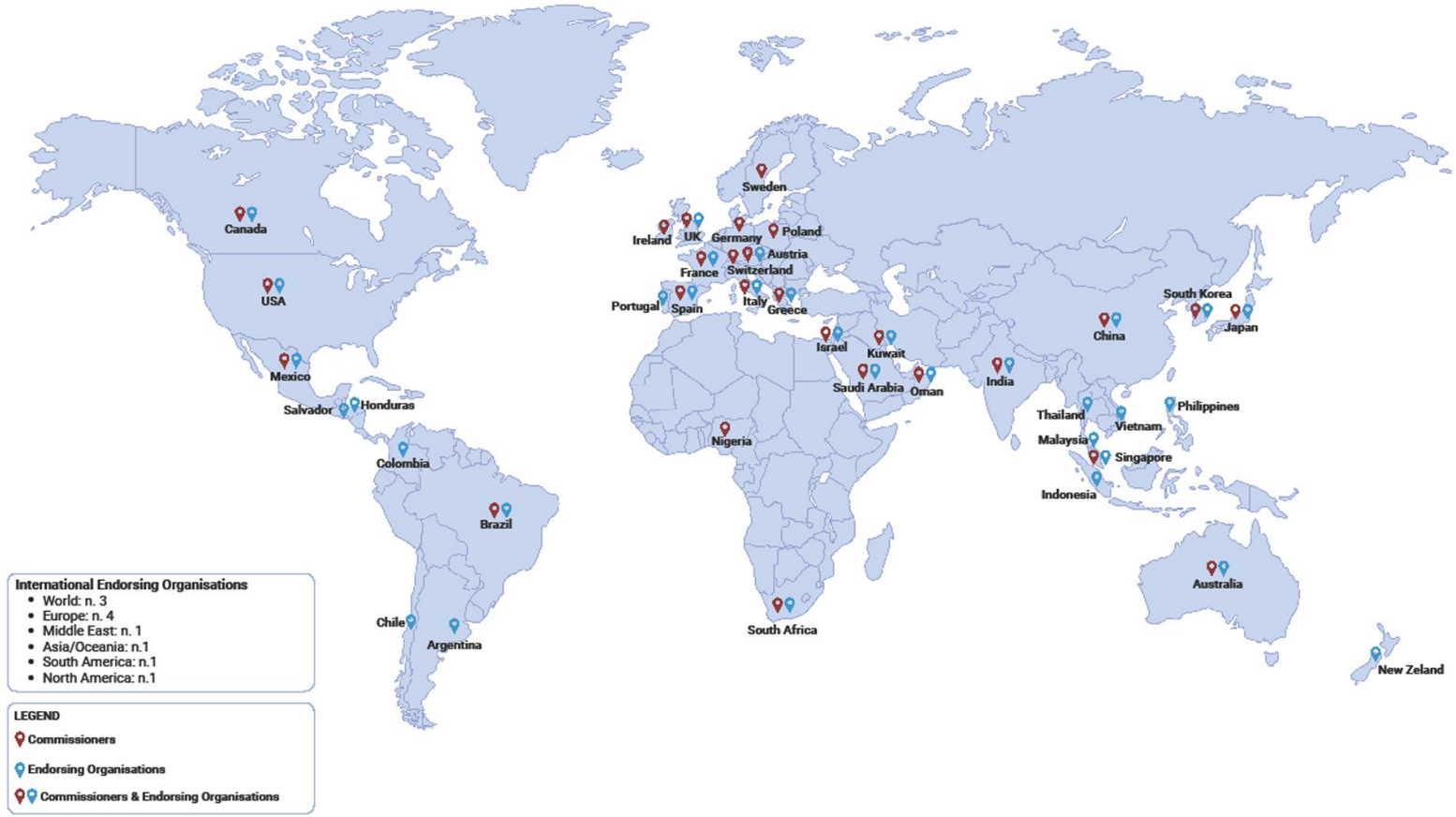
Adapted from Sharma AM. Obes Rev 2010;11:808–9; Guh DP et al. BMC Public Health 2009;9:88; Luppino FS et al. Arch Gen Psychiatry 2010;67:220–9; Simon GE et al. Arch Gen Psychiatry 2006;63:824–30; Church TS et al. Gastroenterology 2006;130:2023–30; Li C et al. Prev Med 2010;51:18–23; Hosler AS. Prev Chronic Dis 2009;6:A48; Lammert F et al. Nat Rev Dis Primers 2016;2:16024; Powell-Wiley TM et al. Circulation 2021;143:e984–1010; Larsson SC et al. Metabolism 2022;137:155326.

New Concept: "Clinical Obesity"

- **The Lancet Commission on Obesity**
- เสนอแนวคิด "**Clinical Obesity": วินิจฉัยเมื่อมี ไขมันส่วนเกิน** (Adiposity) **ร่วมกับ** ความผิดปกติของเนื้อเยื่อ/อวัยวะ หรือมีภาวะแทรกซ้อนที่เกิดจากไขมันส่วนเกินแล้ว (เช่น หายใจลำบาก, ปวดข้อ, ภาวะ metabolic dysfunction)
- **Key**: การวินิจฉัย **ไม่จำเป็นต้องยึดติดกับ BMI เพียงอย่างเดียว** แต่ต้องประเมินผลกระทบทางคลินิกที่ชัดเจน

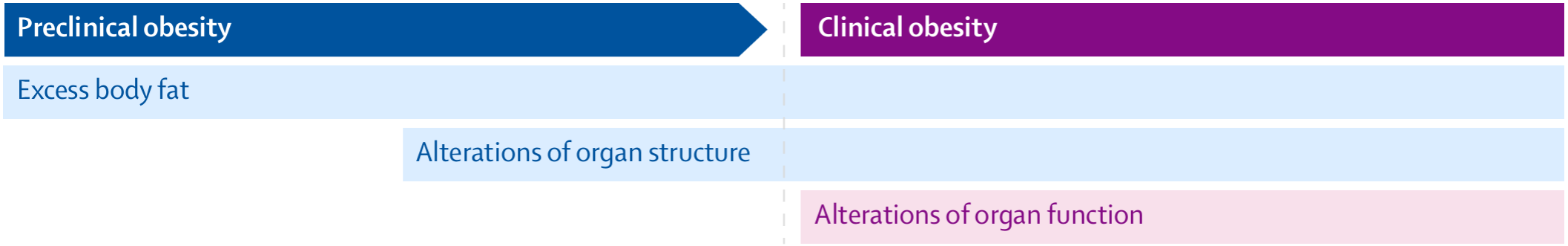
Rubino, Francesco et al. The Lancet Diabetes & Endocrinology, Volume 13, Issue 3, 221 - 262

58 commissioners' expertise
79 organizations endorsed the consensus statements



Redefines obesity as a spectrum of disease states

The pathophysiology of preclinical and clinical obesity



Preclinical obesity

A condition of excess body fat associated with variable level of health risk, but no ongoing illness

People living with preclinical obesity:

- Have no evidence of reduced organ or tissue function due to obesity
- Can complete day-to-day activities unhindered
- Are generally at a higher risk of developing diseases, such as:
  - Clinical obesity
  - Cardiovascular disease
  - Some cancers
  - Type 2 diabetes

Clinical obesity

A chronic disease due to obesity alone, and characterised by signs and symptoms of ongoing organ dysfunction and/or reduced ability to conduct daily activities

People living with clinical obesity have reduced tissue or organ function due to obesity, such as:

- Breathlessness caused by effects of obesity on the heart or lungs
- Knee or hip pain with joint stifness and reduced range of motion
- A cluster of metabolic abnormalities
- Dysfunction of other organs including kidneys, upper airways, nervous, urinary, and reproductive systems.

Full details of these new categories can be found in the Commission report

Elevated body weight or BMI

Clinical confirmation of abnormal fat mass
(by anthropometric measures or direct body fat measurement)

No → **No obesity** (no excess fat mass)



(eg, athletes)

Yes → **Obesity** (excess fat mass)

Assessment for clinical obesity
Obesity-related organ dyfunction, limitations of daily activities, or both

No → **Preclinical obesity**

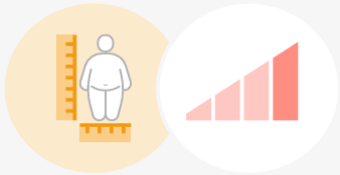


Yes → **Clinical obesity**

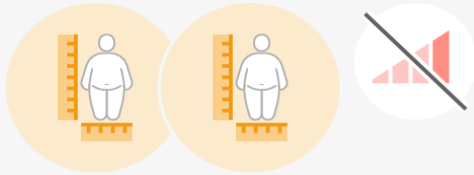


1 Excess body fat?

The first step in such a diagnosis is confirming excess body fat, which can be achieved via one of the following three criteria:



At least one measurement of body size and BMI



At least two measurements of body size, regardless of BMI



Direct body fat measurement, such as a DEXA scan

Male >25%
Female >30%

Measurements of body size

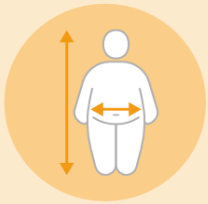
The commission defines three measurements of body size that can be used to confirm excess body fat:



Waist circumference
Male >90 cm
Female >80 cm



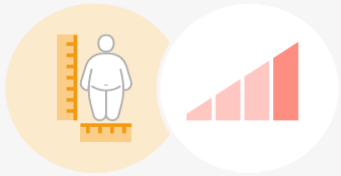
Waist-to-hip ratio
>0.90 for men*
>0.85 for women*



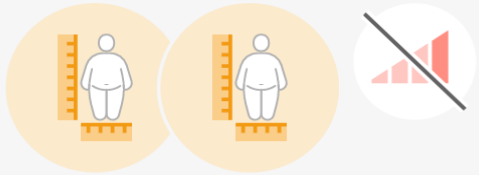
Waist-to-height ratio
>0.50 for all*

Excess body fat can pragmatically be assumed if BMI is >40 kg/m²

1 Excess body fat?
The first step in such a diagnosis is confirming excess body fat, which can be achieved via one of the following three criteria:



At least one measurement of body size and BMI



At least two measurements of body size, regardless of BMI



Direct body fat measurement, such as a DEXA scan

No → **2 No obesity**



Yes → **3 Obesity**
Confirmation of excess body fat should be followed by further assessment to establish whether an individual with obesity has an illness



4 Medical history, physical examination, and standard blood test
- Are there signs/symptoms of organ dysfunction?
- Does the individual experience limitations of day-to-day activities?

No signs/symptoms and no limitations → **Preclinical obesity**

Yes, signs/symptoms of organ dysfunction → **5 Is organ dysfunction obesity related?**

Yes, limitations of day-to-day activities → **Clinical obesity**

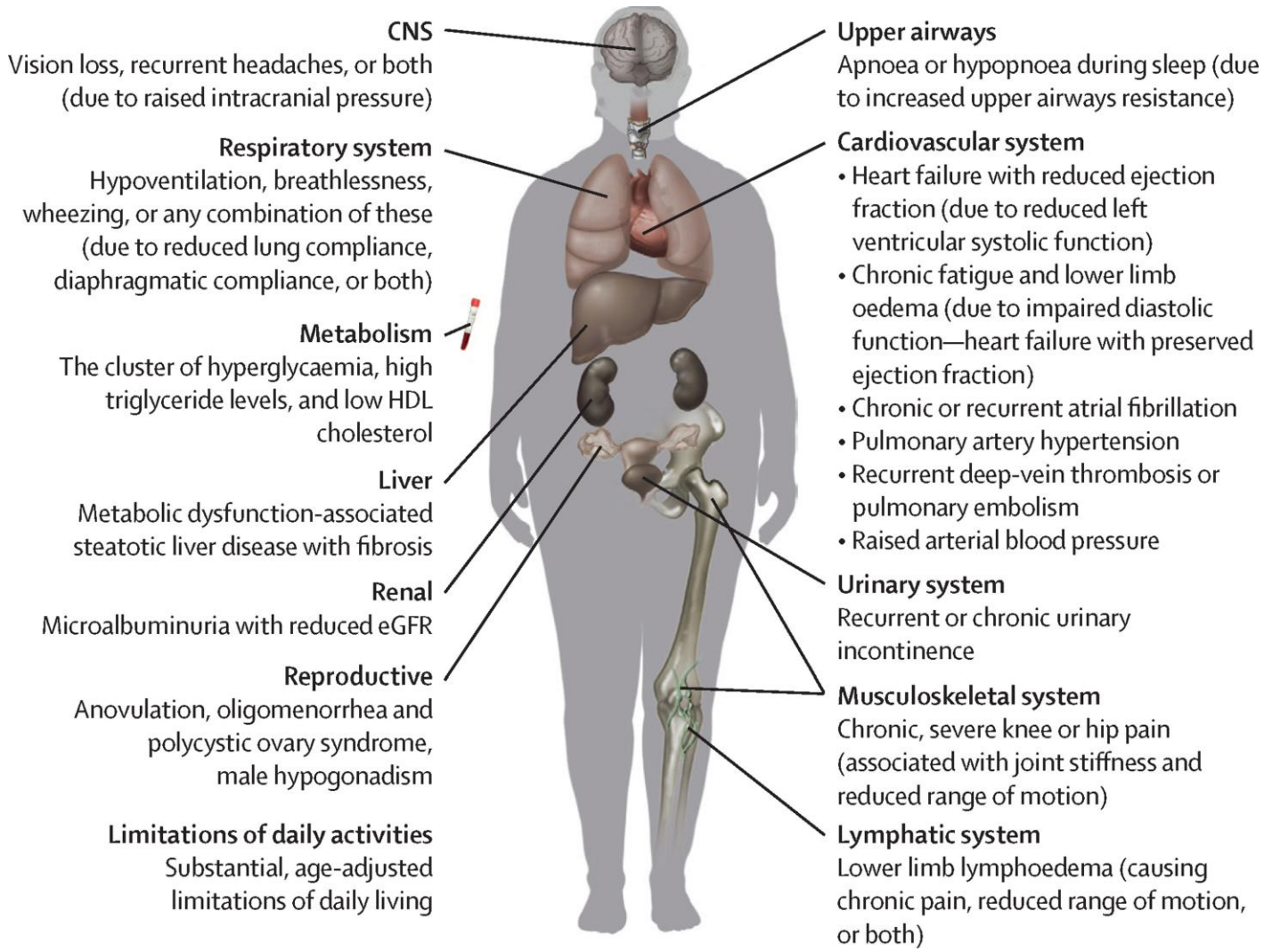
5 Is organ dysfunction obesity related?
- No → **Preclinical obesity**
- Yes → **Clinical obesity**





1

Evidence of obesity related organ/tissue dysfunction



Standard blood test
- CBC
- FPG, Lipid profile
- LFT, Renal function

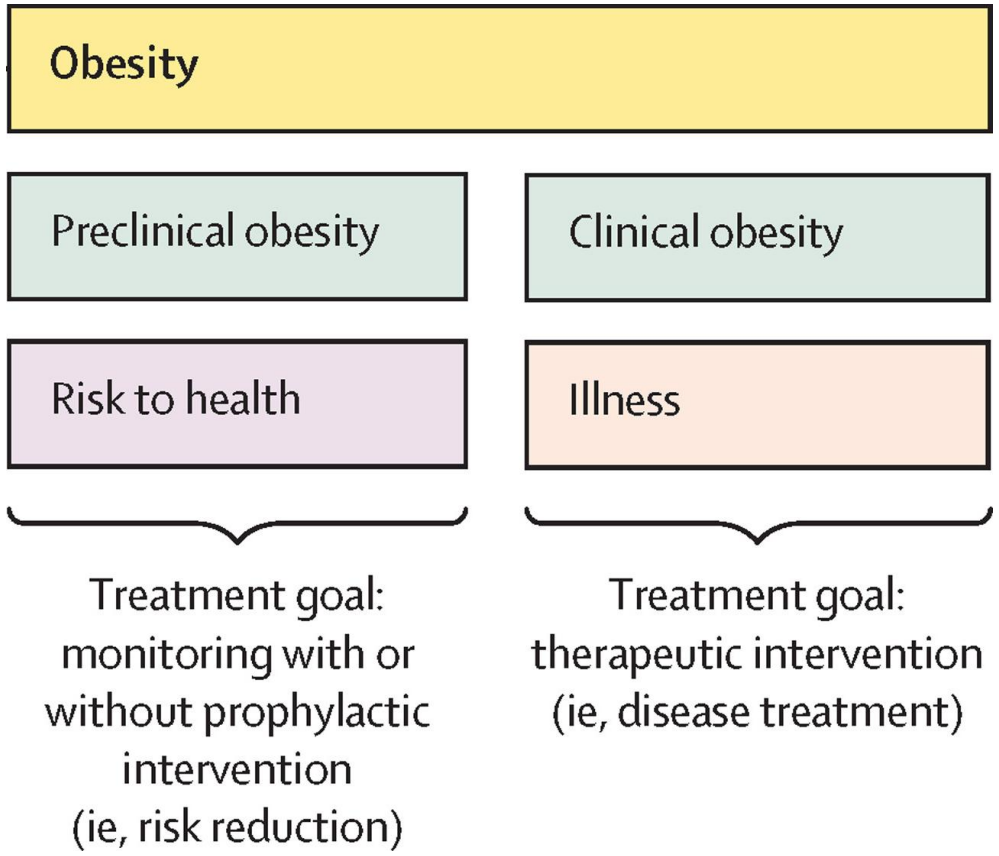
Additional dx test as appropriate

Or…

2

Substantial limitation of mobility and/or other BADLs

Goal of treatment



Preclinical obesity management

Focus on risk reduction and prevention of progression to clinical obesity or other obesity-related diseases

- Health counselling for weight loss or prevention of weight gain
- Monitoring over time
- Active weight loss interventions in people at higher risk of developing clinical obesity, and other obesity-related diseases

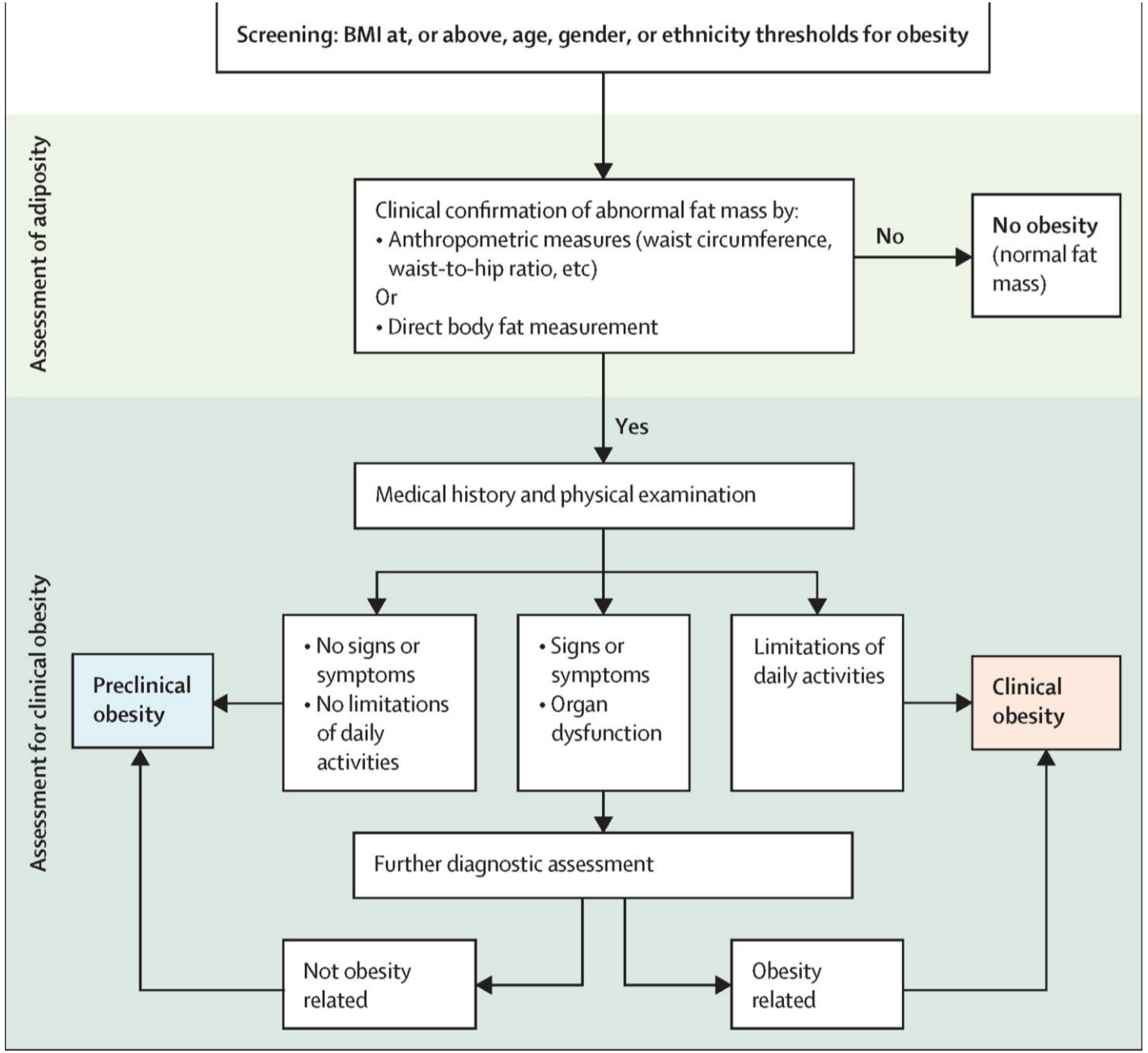
Clinical obesity management

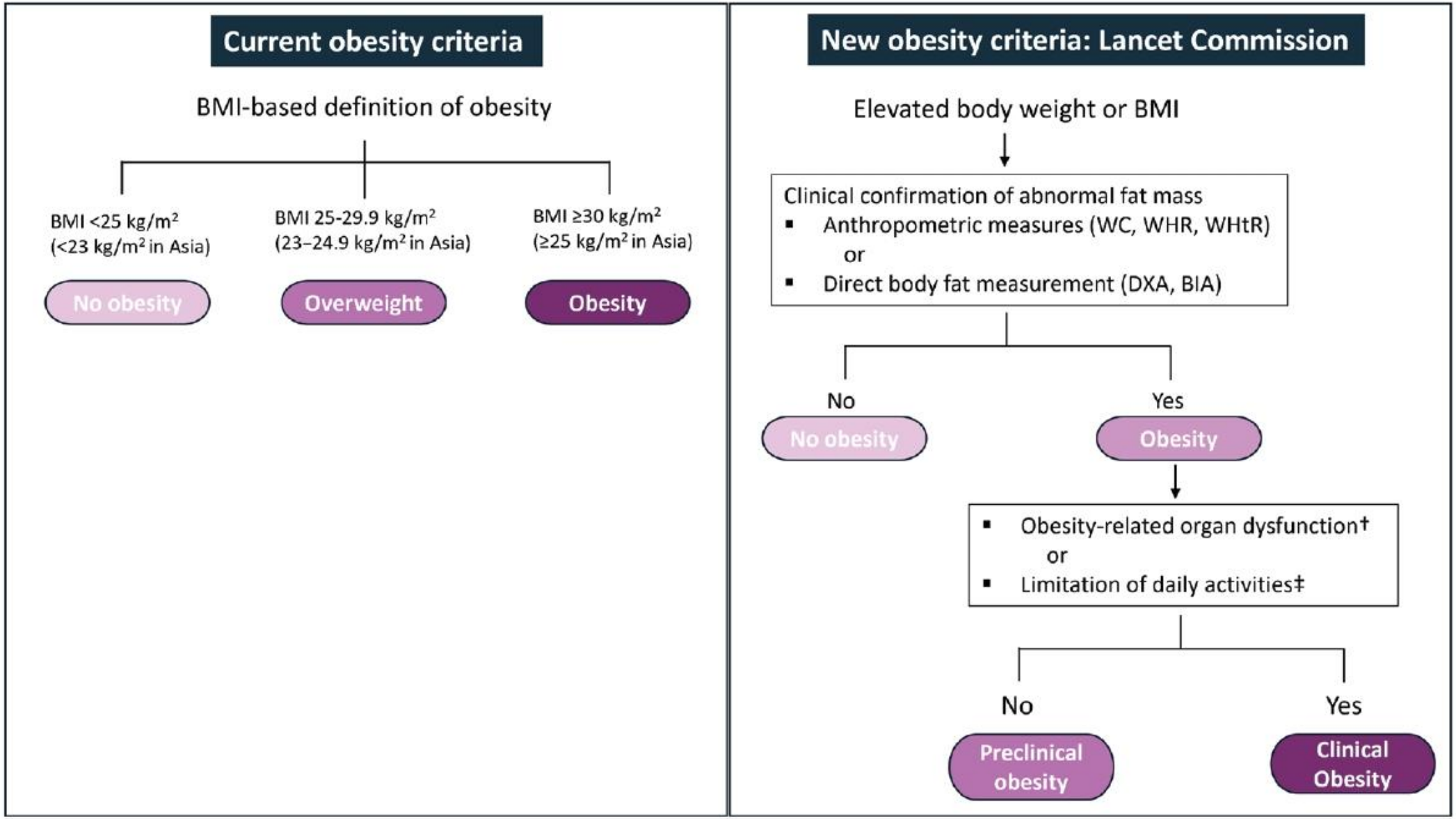
Focus on improvement or reversal of organ dysfunction

- Evidence-based treatment and management, with an aim to fully regain or improve functions
- Treatment type should be informed by individual risk—benefit assessments and decided via an active discussion with the patient
- Success should be assessed by improvement of signs and symptoms, rather than measures of weight loss

Diagnosing clinical obesity

- **BMI is …**
  - useful for screening and epidemiological study
  - not a direct measure of fat,
  - cannot determine when excess body fat is a health problem.
- **Anthropometric criterion**
  - confirmation of excess body fat by anthropometry or direct measurement
- **Clinical criterion for clinical obesity**
  - evidence of compromised organ functions



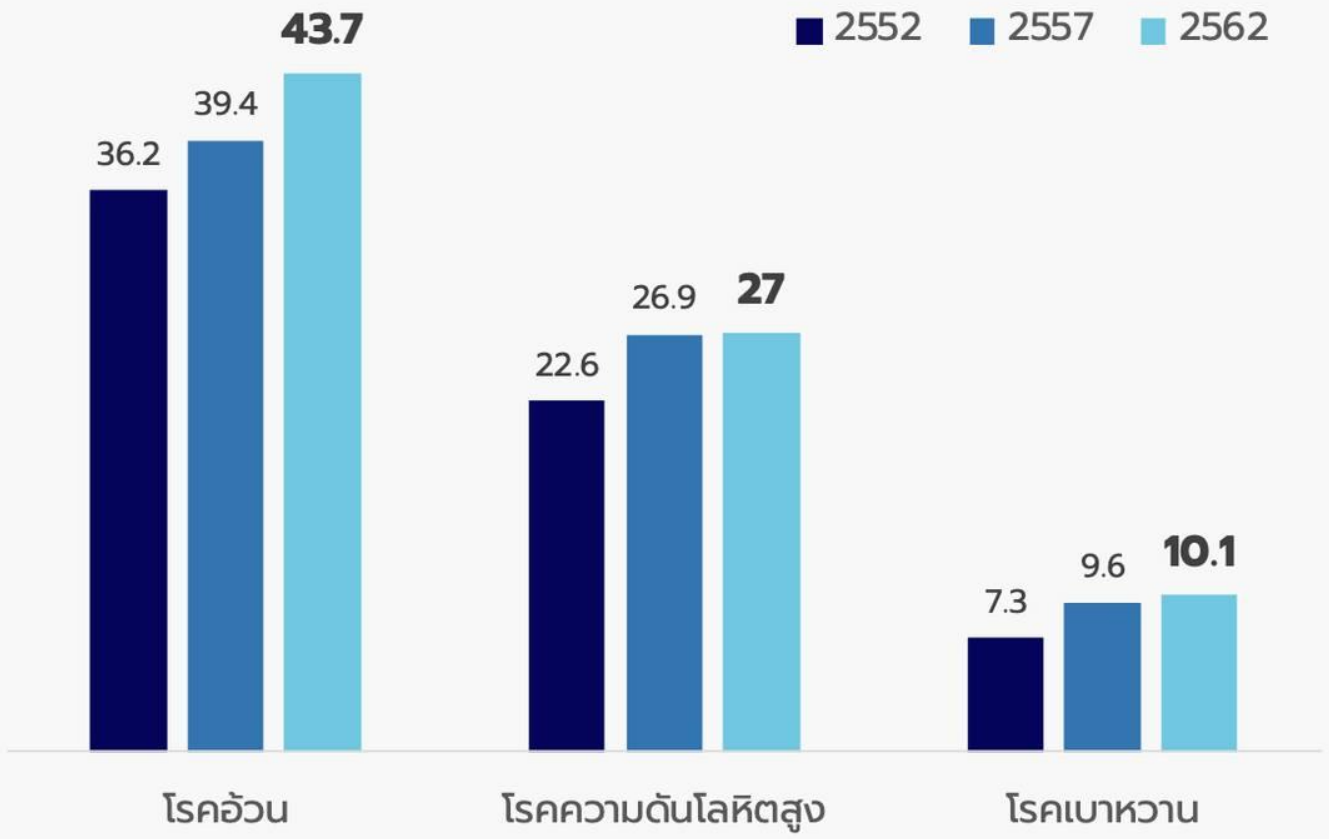

Current obesity criteria
BMI-based definition of obesity
BMI <25 kg/m²
(<23 kg/m² in Asia)
BMI 25-29.9 kg/m²
(23–24.9 kg/m² in Asia)
BMI ≥30 kg/m²
(≥25 kg/m² in Asia)
No obesity
Overweight
Obesity
New obesity criteria: Lancet Commission
Elevated body weight or BMI
Clinical confirmation of abnormal fat mass
Anthropometric measures (WC, WHR, WHtR)
or
Direct body fat measurement (DXA, BIA)
No
Yes
No obesity
Obesity
Obesity-related organ dysfunction†
or
Limitation of daily activities‡
No
Yes
Preclinical obesity
Clinical Obesity

“Obesity affects over 988 million people globally”

คนไทย 1 ใน 2 คน “อ้วน”

อัตราร้อยละการป่วยด้วยโรคเบาหวาน โรคอ้วน โรคความดันโลหิตสูง

(ประชากรไทยอายุ 18 ปีขึ้นไป) พ.ศ. 2552, 2557, 2562



อัตราการเปลี่ยนแปลงในรอบ 10 ปี เพิ่มขึ้น

(เมื่อเทียบกับปี 2552)

- DM 37.9%
- อ้วน 20.7%
- HT 19.5%

แหล่งข้อมูล: รายงานการสำรวจสุขภาพประชาชนไทย โดยการตรวจร่างกาย ครั้งที่ 2552, 2557, 2562

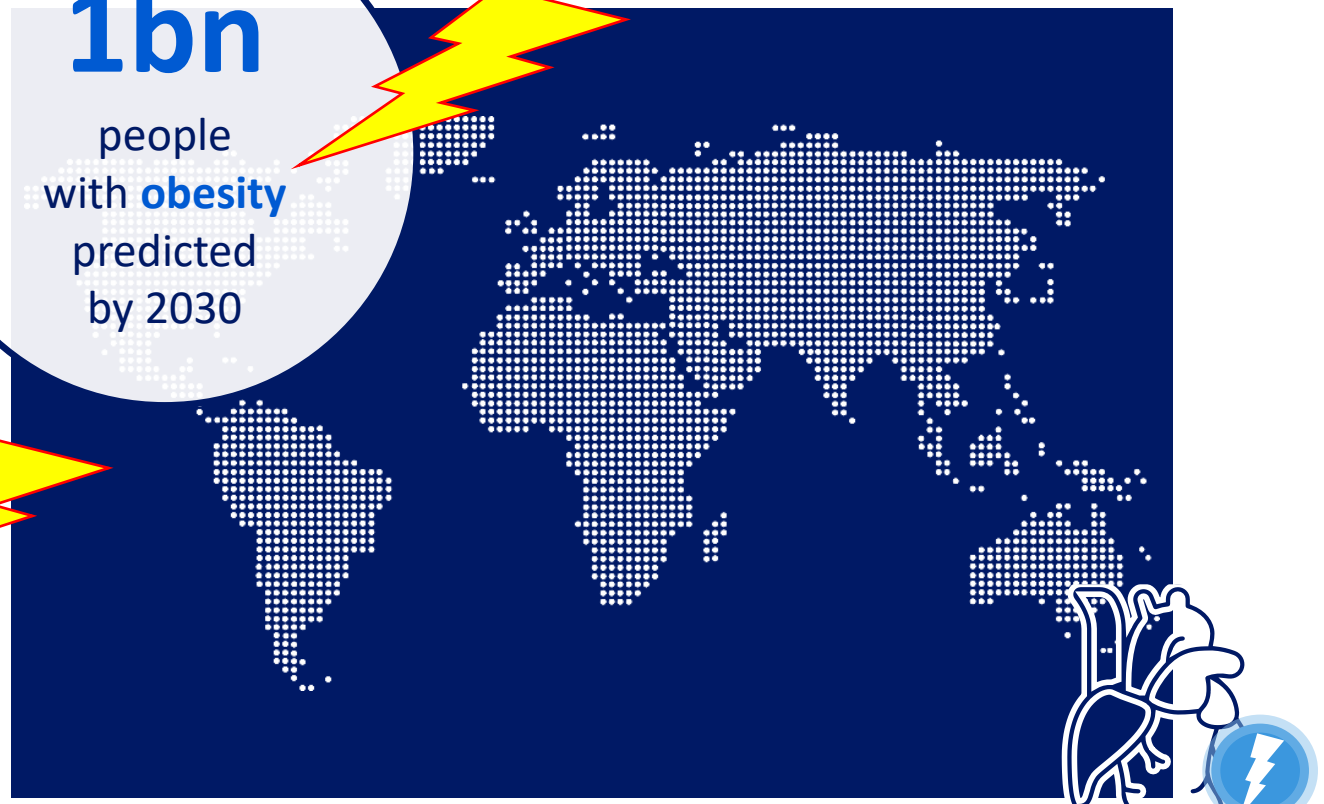
Global prevalence of obesity and CVD

In 2015,
4 million
deaths were due to high BMI and
2/3rd due to
CVD

32%
of all global **deaths** are from **CVD**

523 million
people had **CVD** in 2019 (globally)

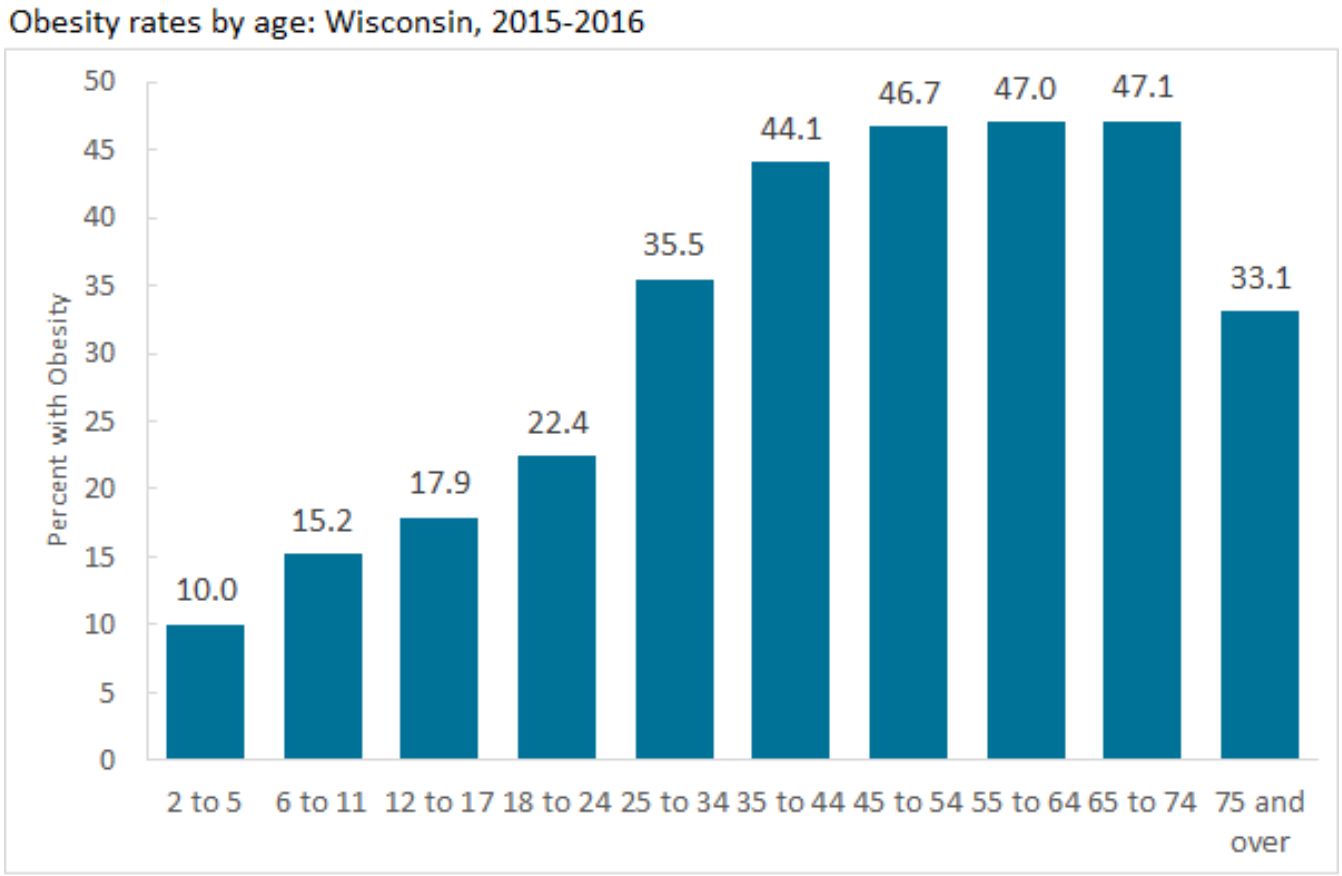
1bn
people with **obesity** predicted by 2030



Bn, billion; BMI, body mass index; CVD, cardiovascular disease.

1. WHO fact sheets. 2021; 2. World Obesity Atlas. 2022; 3. World Obesity Federation; 4. Wiley et al. Circulation. 2021;143:e984-1010.

ยิ่งแก่ ยิ่งอ้วน


Obesity rates by age: Wisconsin, 2015-2016
Percent with Obesity
50
45
40
35
30
25
20
15
10
5
0
10.0
15.2
17.9
22.4
35.5
44.1
46.7
47.0
47.1
33.1
2 to 5
6 to 11
12 to 17
18 to 24
25 to 34
35 to 44
45 to 54
55 to 64
65 to 74
75 and over
Obesity rates were not significantly different among the 45-54, 55-64, and 65-74 age groups.
Based on data from Wisconsin Collaborative for Healthcare Quality, 2015-2016.



1 Many paths can lead to obesity

2 Adaptive biological responses lead to persistence of obesity as a chronic disease

Is obesity only a result of voluntary excessive energy intake and decreased energy expenditure?

Biological
The brain controls **eating behavior and appetite[1].**

Genetic
Around **40-70%** of our weight is dependent on our **genetic make-up[2]**

Psychological
Stress and psychological distress **impact appetite[3]**



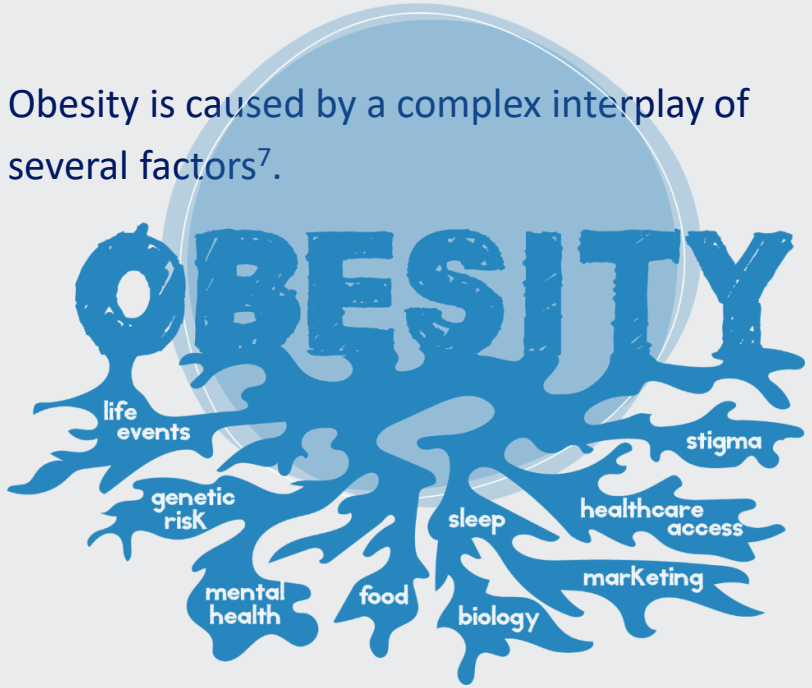
Social
Health, social and economic inequalities[5]

Medications
Several medications **promote long term weight gain[6]**

Environmental
Unhealthy eating patterns and **sedentary lifestyle[2]**

Many paths can lead to obesity

Obesity is caused by a complex interplay of several factors[7].



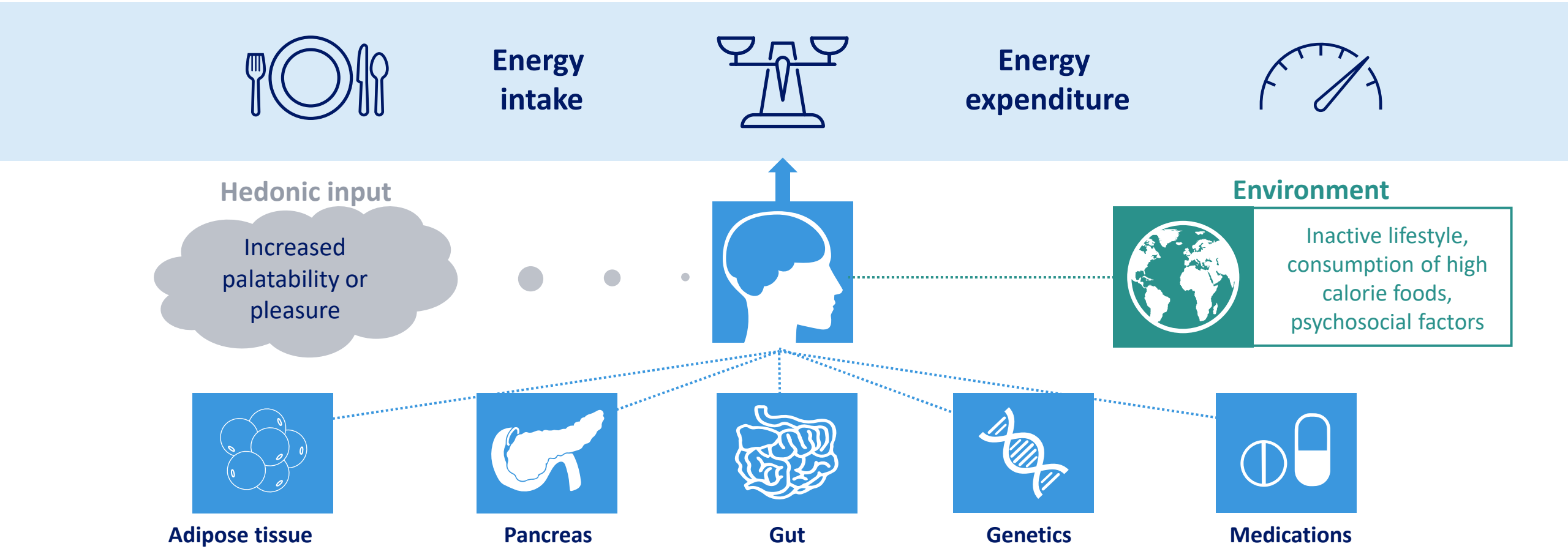
Obesity is not simply due to an individual's choice or lack of willpower[7]

Appetite is normally regulated by a complex interplay between gut and brain[7]

Many paths lead to obesity

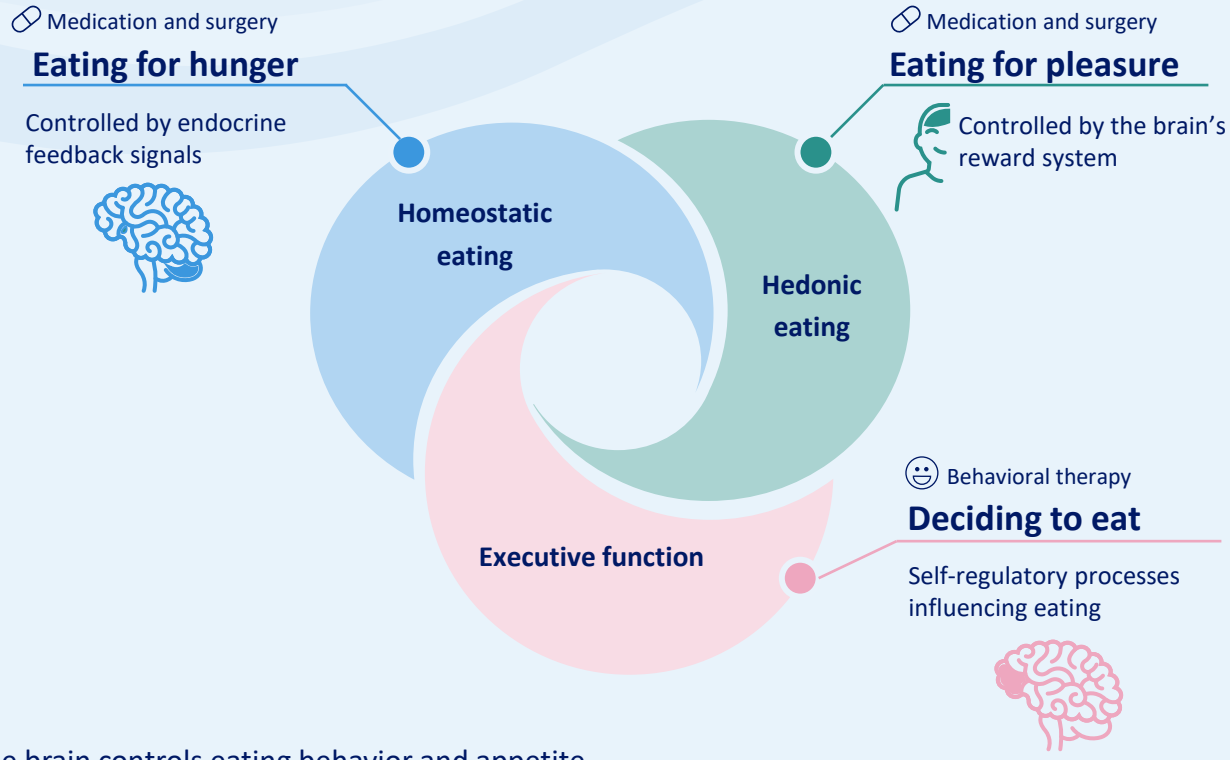
1. Hall KD et al. Am J Clin Nutr. 2022;115:1243–54; 2. Herrera BM and Lindgren CM. Curr Diab Rep. 2010;10:498–505; 3. Sominsky L and Spencer SJ. Front Psychol. 2014;5:434; 4. Luppino FS et al. Arch Gen Psychiatry. 2010;67:220–9; 5. Loring & Robertson. 2014. The Regional Office for Europe of the World Health Organization, pp.4; 6. Janssen LK & Juk JL. Canadian Adult Obesity. Available from https://obesitycanada.ca/guidelines/epidemiology. Accessed July 2022; 7. Lau D et al. Canadian Adult Obesity Clinical Practice Guidelines: The Science of Obesity. Available from https://obesitycanada.ca/guidelines/science. Accessed November 2022.

Energy balance is regulated by the brain through various sources of input



1. González H. Managing Patients with Obesity. Springer International Publishing Switzerland; 2016;
2. Gadde et al. J Am Coll Cardiol. 2018;71:69–84; 3. Heymsfield & Wadden. N Engl J Med. 2017;376:254–66.

What is the role of the brain in regulating appetite?



The brain controls eating behavior and appetite. Weight is determined and regulated by a unique, three-layer appetite system.

Appetite is normally regulated by a complex interplay of different signals and areas in the brain

Increased understanding of the biology of appetite regulation has led to the development of new generation pharmacotherapy.

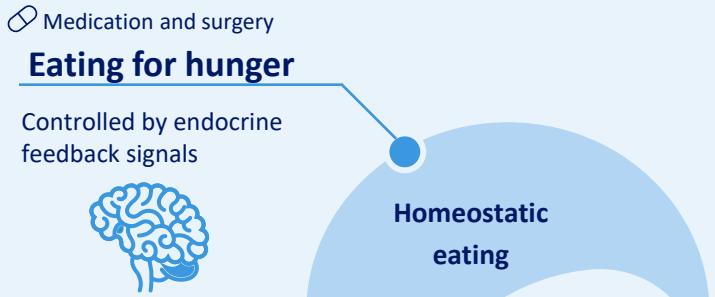
Many paths lead to obesity

Obesity is not simply due to an individual's choice or lack of willpower[7]

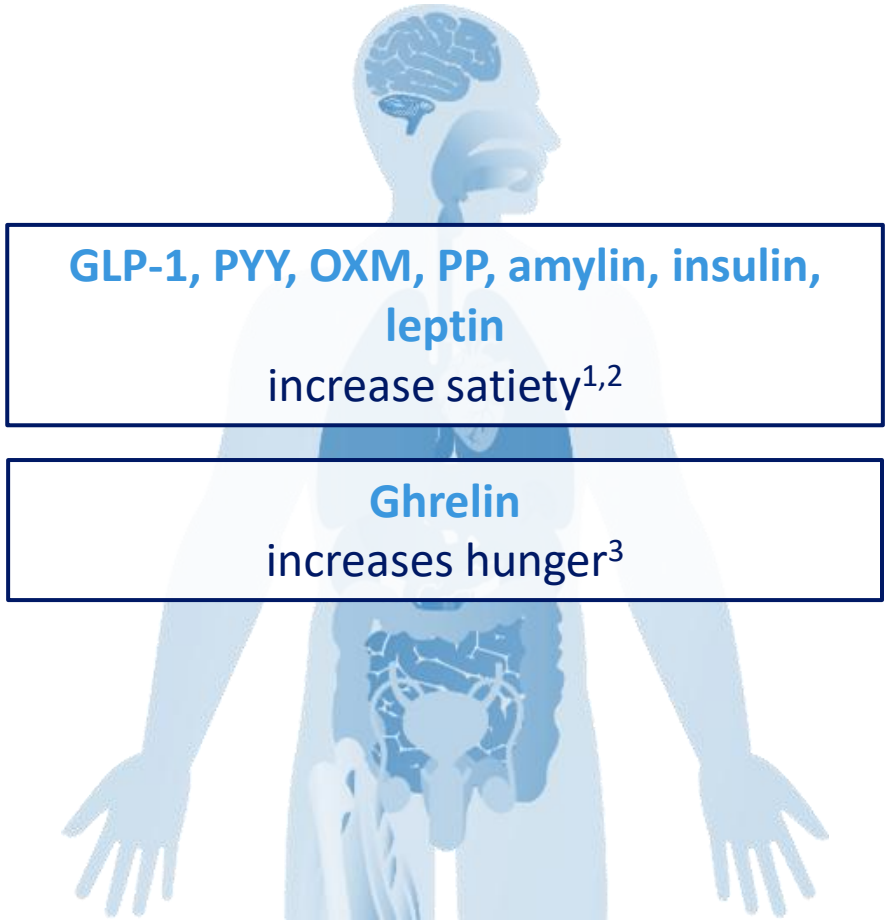
Appetite is normally regulated by a complex interplay between gut and brain

1. Badman MK & Flier JS. Science. 2005;307:1909–14; 2. van Bloemendaal L et al. Diabetes. 2014;63:4186–96; 3. Klok MD et al. Obes Rev. 2007;8:21–34; 4. Hall K et al. Am J Public Health. 2014;104:1169–75; 5. Berridge KC et al. Brain Res. 2010;1350:43–64; 6. Vallis M. Clin Obes. 2019;9:e12299; 7. Lau D et al. Canadian Adult Obesity Clinical Practice Guidelines: The Science of Obesity. Available from https://obesitycanada.ca/guidelines/science.

The role of the brain in controlling appetite



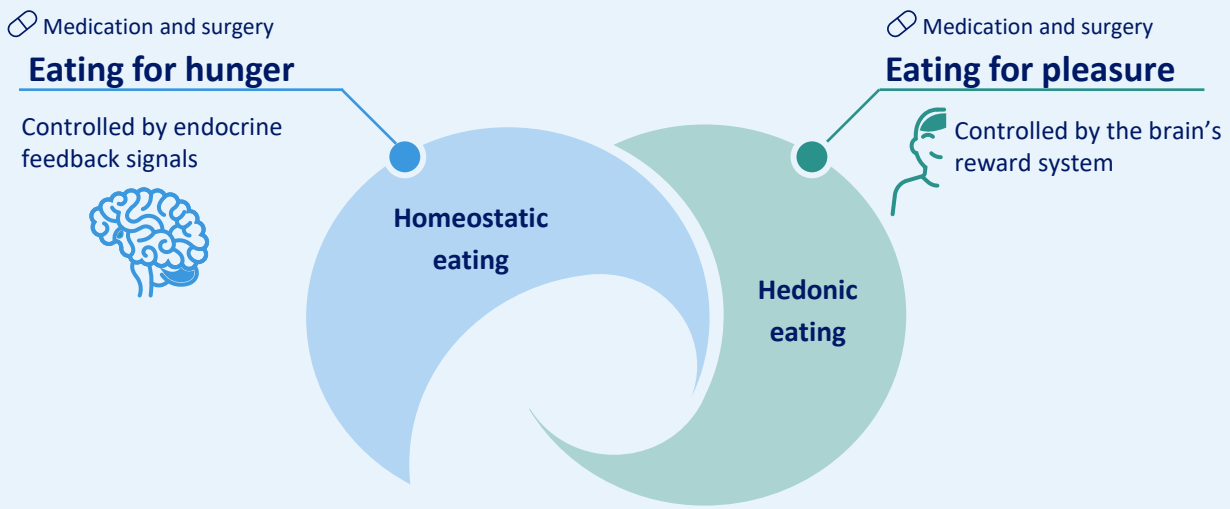
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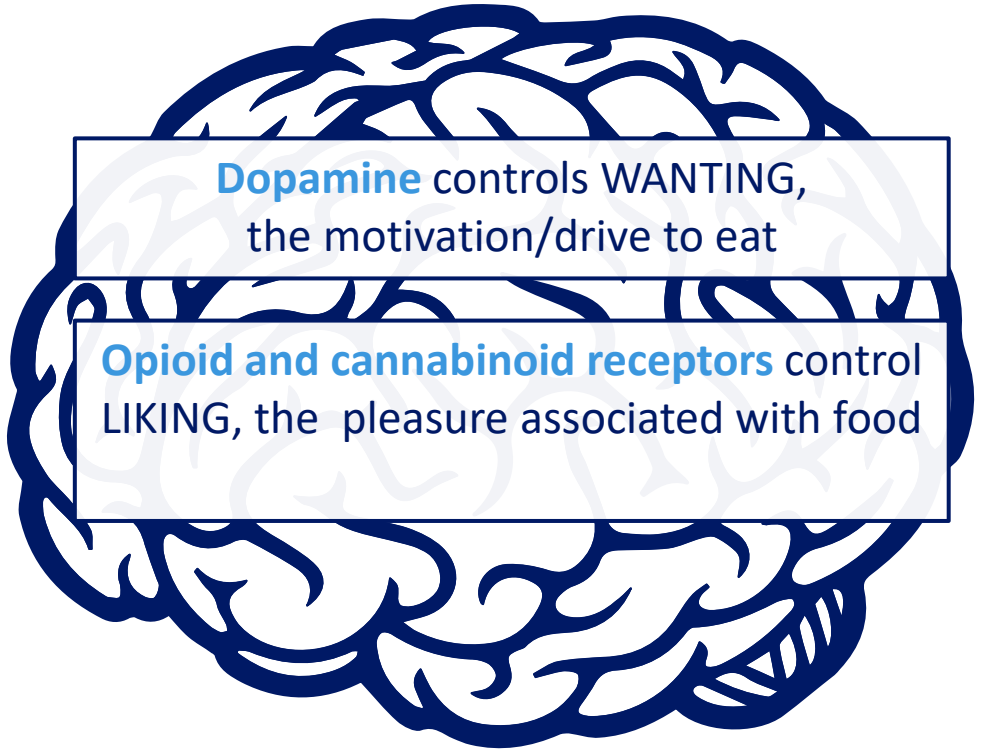
GLP-1, glucagon-like peptide-1; POMC, pro-opiomelanocortin; PP, pancreatic polypeptide; PYY, peptide YY; OXM, oxyntomodulin

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The role of the brain in controlling appetite



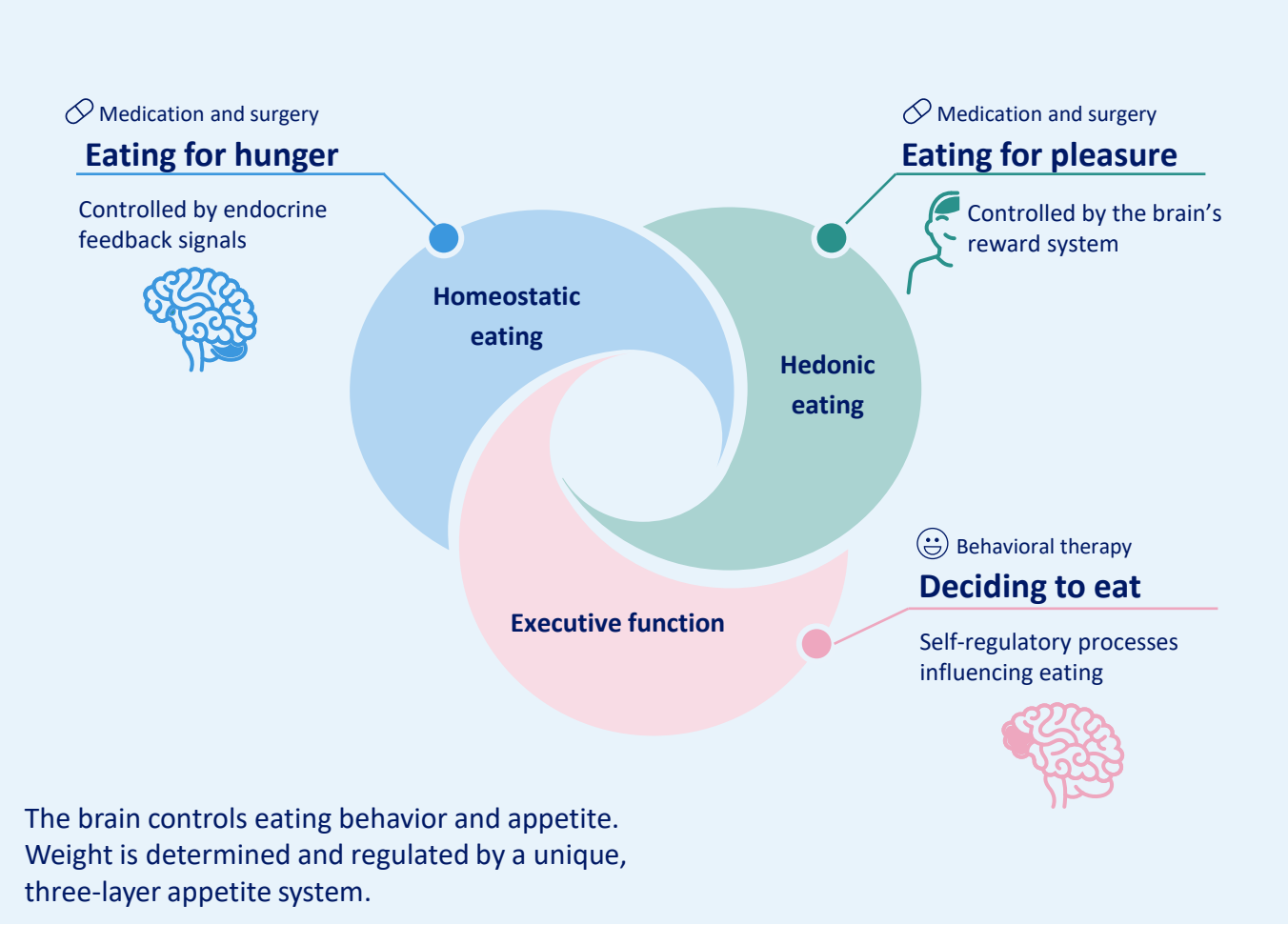
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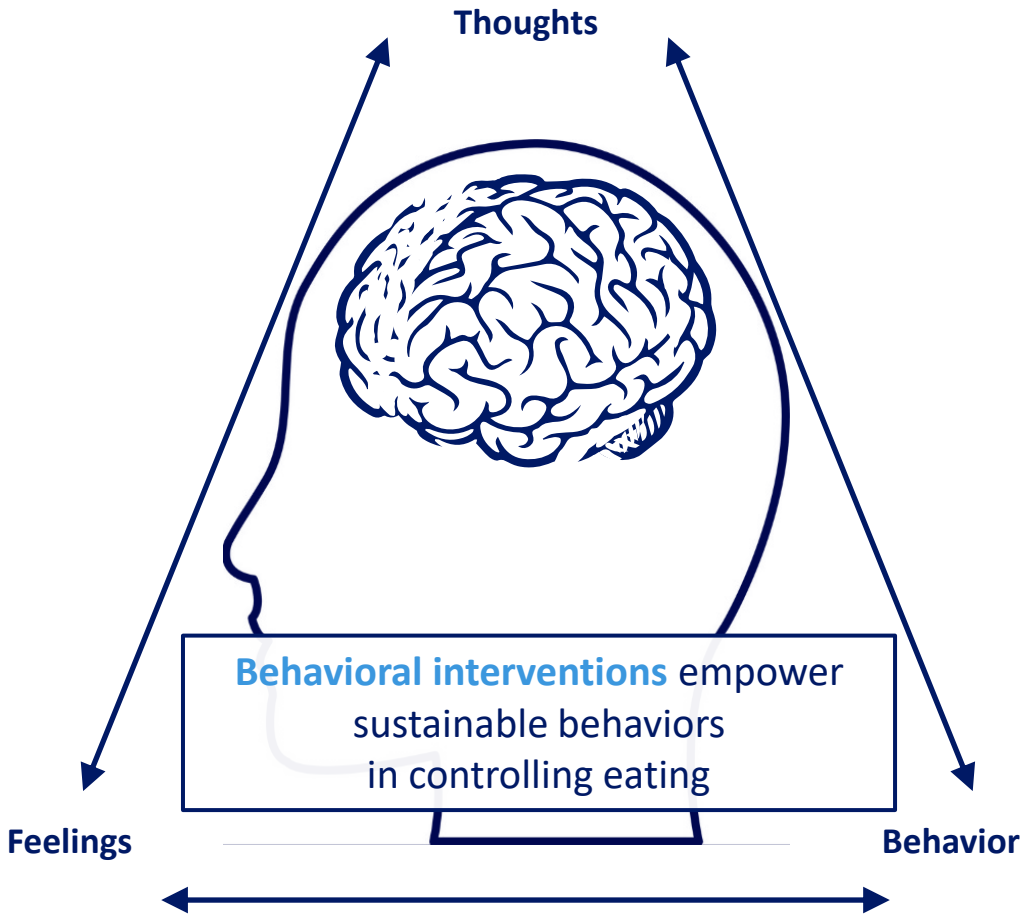
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The role of the brain in controlling appetite



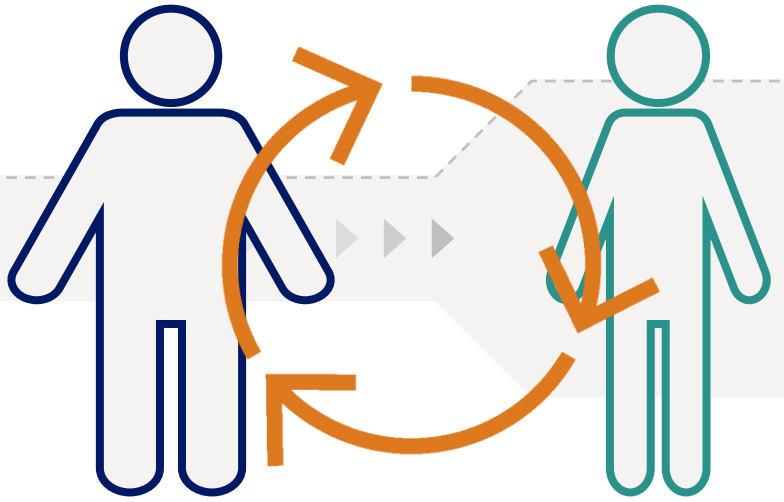
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Weight loss alters the body's homoeostatic system by limiting weight loss and favouring weight regain



Yo-Yo circle

BEFORE WEIGHT LOSS

AFTER WEIGHT LOSS

- ↑ Hunger
- Prefer calorie dense foods

ENERGY GAP

- ↓ Energy expenditure

Metabolism:

- ↓ Fat oxidation
- ↑ Cortisol

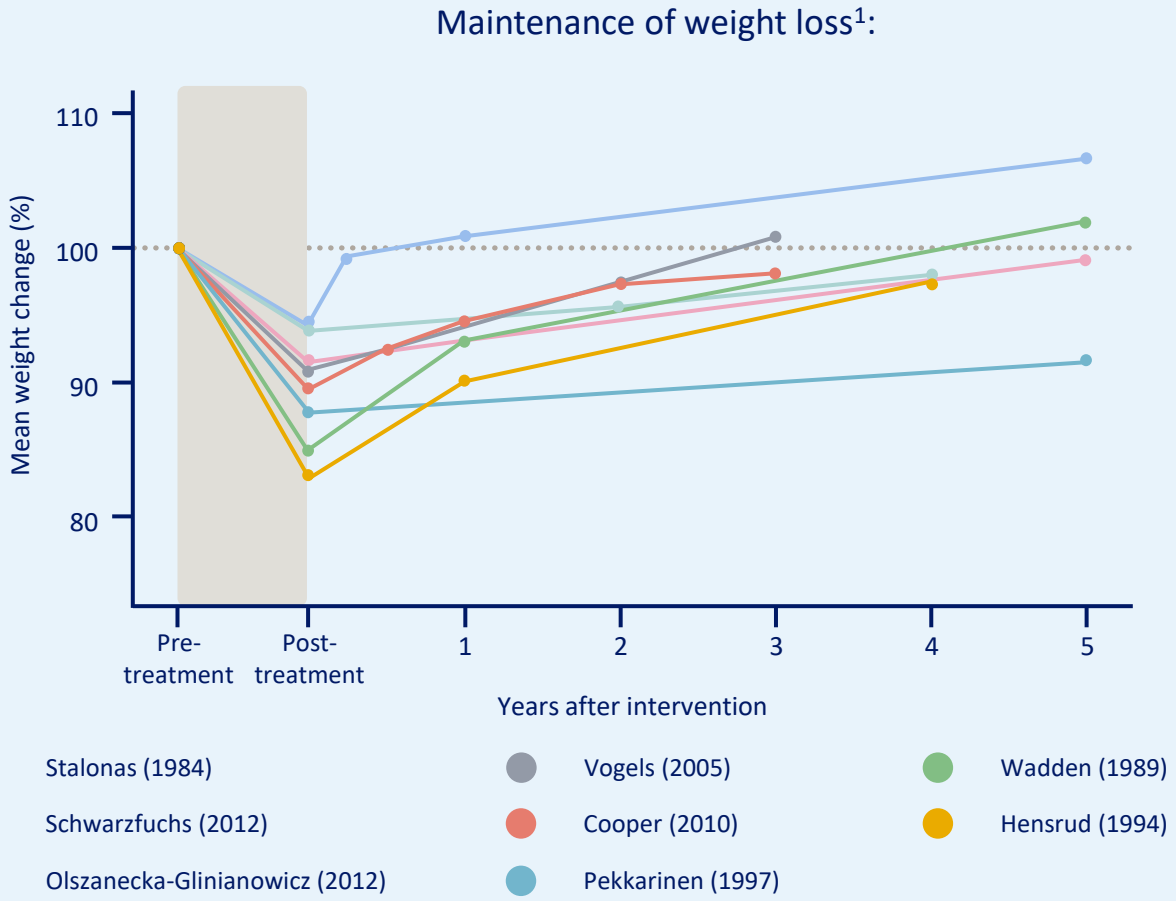
Hormone levels

- ↓ Satiety hormones (leptin, POMC, PYY, GLP-1, CART)
- ↑ Hunger hormones (ghrelin, NPY, AgRP)
- ↓ Nervous system activity
- ↓ Body Mass
- ↓ Leptin
- ↑ Energy Efficiency

AgRP, agouti-related peptide; CART, cocaine-amphetamine related transcript; GLP-1, glucagon-like peptide 1; NPY, neuropeptide Y; POMC, pro-opiomelanocortin; PYY, peptide YY.

Melby et al. Nutrients 2017; 9(5); Sumithran et.al J. Clin Sci 2013;124:231–241; Fothergill E et al. Obesity (Silver Spring) 2016;24:1612–1619; Sumithran P et al. N Engl J Med 2011;365:1597–1604.

How successful are lifestyle interventions in chronic weight management?



Weight loss following lifestyle interventions is not sustainable in the long-term

Most individuals regain more than half of the weight lost within 2 years and more than 80% within 5 years[2].

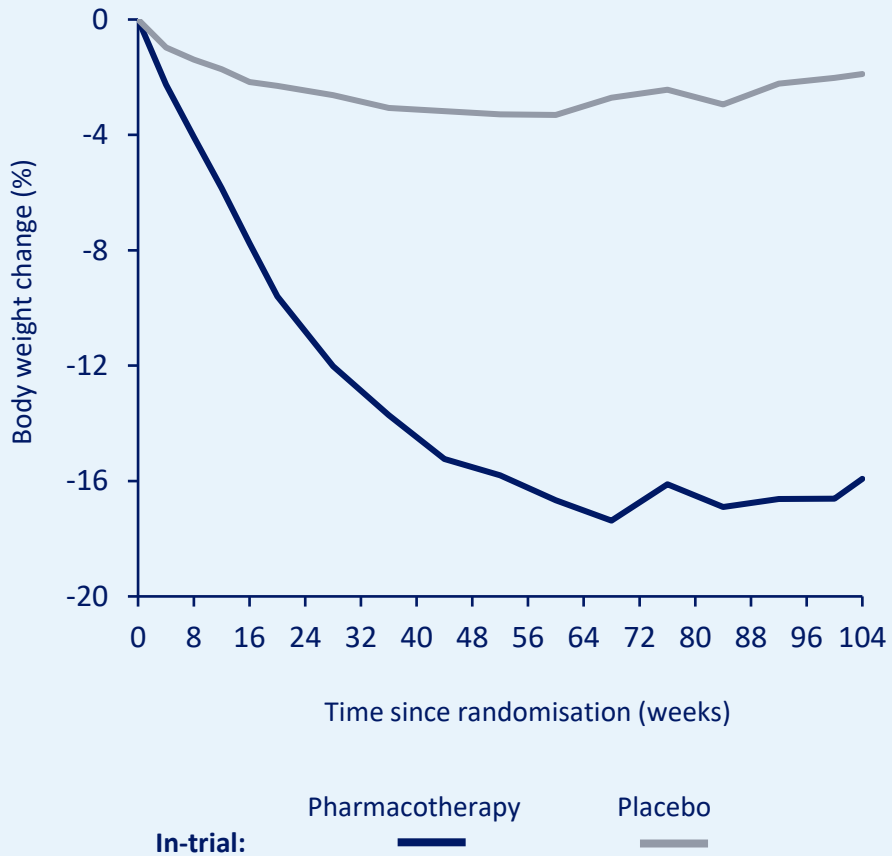
Moreover, lifestyle interventions often result in modest weight loss (3–5%).

Diet & exercise alone <5% weight loss[2]

Sustained weight loss leads to health benefits and improvements in quality of life[3]

1. Nordmo M et al. Obes Rev. 2020;21:e12949; ; 2. Horn D et al. Postgrad Med. 2022;134:359–75.

Why is it recommended to incorporate pharmacotherapy as standard of care for obesity management?



Placebo[1]

-2.6%
body weight change

with diet and exercise alone

Pharmacotherapy[1]

-15.2%
body weight change

with addition of pharmacotherapy

Long-term pharmacotherapy is an integral part of obesity management

New generation of anti obesity medications can:

- Maximize the magnitude of weight loss
- Help to overcome the compensatory mechanisms that induce weight regain, promoting sustained weight loss

Diet & exercise alone
<5% weight loss[2]

New generation pharmacotherapy
15--17% weight loss[2-7]

Maintenance of weight loss

1. Garvey WT et al. Nat med 2022;28:2083-91; 2. Wadden et al. N Engl J Med 2005;353:2111–20; 3. Wilding et al. N Engl J Med 2021;384:989-1002; 4. Torgerson J. S., et al. Diabetes care 2004;27:155-61; 5. Apovian C., et al. Obesity 2013;21:935-43; 6. Pi-Sunyer, X., et al. N Engl J Med 2015;373:11-22; 7. Allison D. B., et al. Obesity 2012;20:330-42;
8. Horn D et al. Postgrad med 2022;134:359-75.

Obesity is recognised as a disease and a health issue

"Obesity is a chronic, relapsing, progressive disease process ….need for immediate action for prevention and control of this global epidemic"

World Obesity Federation[1]

"Obesity is a progressive chronic disease, similar to diabetes or high blood pressure, …"

Obesity Canada[3]

"A progressive disease, impacting severely on individuals and society alike,… obesity is the gateway to many other disease areas…"

European Association for the Study of Obesity[4]

"Obesity and overweight as a chronic medical condition (de facto disease state) and urgent public health problem…"

American Medical Association[2]

"It (obesity) is not a lifestyle choice caused by individual greed but a disease caused by health inequalities, genetic influences and social factors.."

Royal College of Physicians UK[5]

"The Treat and Reduce Obesity Act would allow a variety of qualified practitioners, including registered dietitian nutritionists, to more effectively treat this disease, which impacts more than one-third of our nation."

Academy of nutrition and dietetics[6]

"Obesity is a chronic relapsing disease, which in turn acts as a gateway to a range of other non-communicable diseases, such as diabetes, cardiovascular diseases and cancer."[3]

European Commission[7]

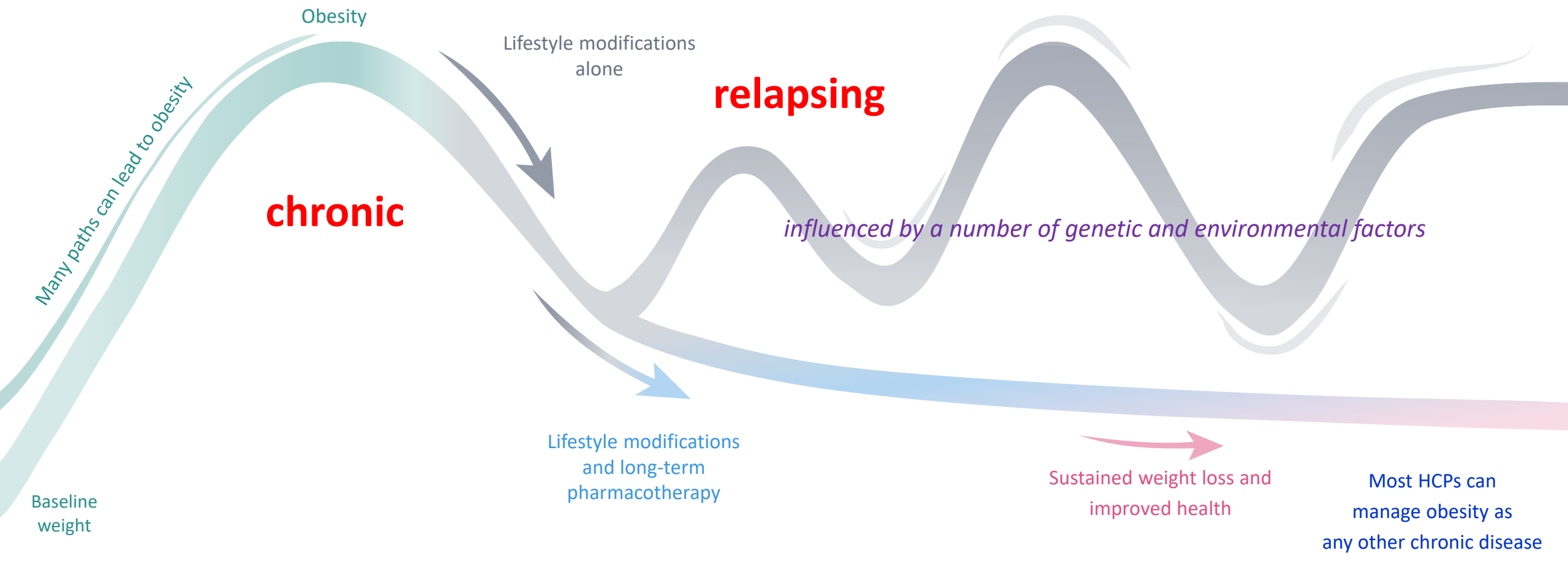
"A pathological state (obesity disease) in which a person suffers health problems caused by or related to obesity thus making weight loss clinically desirable …"

Asia Oceania Association for the Study of Obesity[8]

1. Bray et al. Obes Rev 2017;18:715–23; 2. AMA resolutions. June 2012. Available at https://www.ama-assn.org/sites/ama-assn.org/files/corp/media-browser/public/hod/a12-resolutions_0.pdf. Accessed March 2024; 3. Obesity Canada. Available at https://obesitycanada.ca/guidelines/. Accessed March 2024; 4. EASO: 2015 Milan Declaration: A Call to Action on Obesity. Available at https://easo.org/2015-milan-declaration-a-call-to-action-on-obesity/. Accessed March 2024; 5. Royal College of Physicians. Anon. BMJ 2019;364:l45; 6. Raynor et al. J Acad Nutr Diet 2016;116:129–47; 7. European Commission. Obesity prevention. Available from https://knowledge4policy.ec.europa.eu/health-promotion-knowledge-gateway/obesity_en. Accessed March 2024; 8. AOASO position statement, Nagoya Declaration 2015.

Obesity is a…

progressive disease.



requires safe & effective evidence-based treatment for long-term care